

Raspberry Pi

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General

- This page give a general overview over the Raspberry Pi platform (hardware and software)
- for TRENTOS-M specific information see also
 - [TRENTOS-M on RPi 3](#)
 - [TRENTOS-M on RPi 4](#)

Models

There is a comprehensive description of each model at

- https://en.wikipedia.org/wiki/Raspberry_Pi
- <https://www.raspberrypi.org/documentation/configuration/uart.md>

The main difference that affect us are:

- Model 2B
 - UARTs
 - UART0: PL011, routed to GPIO 14 (Tx, Pin 8) and 15 (Rx, Pin 10)
 - UART1: proprietary mini UART, connected to Bluetooth chip
- Model 3B
 - Broadcom BCM2837
 - 4x Cortex-A53 at 1.2 GHz
 - proprietary GIC
 - UARTs
 - UART0: PL011, connected to Bluetooth chip
 - UART1: proprietary mini UART, routed to GPIO 14 (Tx, Pin 8) and 15 (Rx, Pin 10)
 - 1 GiB DRAM
 - Microchip LAN9514 (sm5c9514)
 - connected to SOC's USB 2.0 port
 - 4 port USB 2.0 hub
 - 10/100 Ethernet
- Model 3B+
 - basically like model 3B
 - new SOC revision BCM2837B0 with 1.4 GHz
 - Microchip LAN7515
 - connected to SOC's USB 2.0 port
 - 4 port USB 2.0 hub
 - 10/100/1000 Ethernet (max throughput is 300 MBit due to USB bandwidth limitation)
 - RUN/PEN pins
 - RUN pulled high by default, pull down to GND to make CPU reset
 - PEN (Power Enable) pulled high by default, pull down to GND to make PMIC shut down and cut power
- Model 4
 - Broadcom BCM2711
 - https://www.raspberrypi.org/documentation/hardware/raspberrypi/bcm2711/rpi_DATA_2711_1p0_preliminary.pdf
 - <https://www.heise.de/ct/artikel/Raspberry-Pi-4-Model-B-Blockschaltbild-des-Broadcom-BCM2711-4514399.html>
 - 4x Cortex-A72, 1.5 GHz
 - "standard" ARM GIC-400 (GICv2)
 - native 10/100/1000 ethernet

- UART
 - UART0: PL011, connected to Bluetooth chip
 - UART1: proprietary mini UART, routed to GPIO 14 (Tx, Pin 8) and 15 (Rx, Pin 10)
 - UART2: PL011 (GPIO 0/1)
 - UART3: PL011 (GPIO 4/5)
 - UART4: PL011 (GPIO 8/9)
 - UART5: PL011 (GPIO 12/13)
 - USB-C power supply
 - 1, 2, 4 or 8 GiB DRAM
- Model 4 v1.2
 - USB-C power connector works according to USB-C specs

Hardware

VideoCore

- <https://github.com/raspberrypi/firmware>

Device ID and serial

- mailbox Command GET_BOARD_SERIAL
 - native (taken from <https://github.com/raspberrypi/firmware/wiki/Mailbox-property-interface>)
 - mailbox 0, channel 8, tag 0x00010004
 - Request Length: 0
 - Response
 - Length: 8
 - Value: u64: board serial
- in Linux it can be found in /proc/cpuinfo:

```
cat /proc/cpuinfo | grep Serial | cut -d ' ' -f 2

[...]
```

Hardware	: BCM2835
Revision	: c03111
Serial	: 1000000003d71d0c
Model	: Raspberry Pi 4 Model B Rev 1.1

PL011 UART

standard ARM peripheral, see <http://infocenter.arm.com/help/topic/com.arm.doc.subset.primecell.peripherals/index.html#pl01x>

mini UART

- small FIFOs
- no DCD, DSR, DTR or RI signals
- lack of flow control: no break detection, framing errors detection, parity bit
- no receive timeout interrupt

RasPi3 proprietary GIC

- ToDo ?

Bootloader

- armstub
 - <https://github.com/seL4/seL4/pull/454>
 - <https://leiradel.github.io/2019/01/20/Raspberry-Pi-Stubs.html#armstub8s>
- U-Boot
 - see also [U-Boot](#)
 - The default baud rate for UART0 specified in the uboot .env is **115200**

seL4 on Raspberry Pi

- see [seL4](#) for a general background
- [SEOS-1229](#) - Getting issue details... STATUS
- Data61 port has been done on RasPi 3B, not 3B+
- seL4 documentation that explains how to set up the SD card: <https://docs.sel4.systems/Hardware/Rpi3.html>
- random link collection
 - <https://research.csiro.au/tsblog/sel4-raspberry-pi-3>
 - <https://inner-haven.net/posts/2017-09-15-sel4-rpi3.html>
 - <http://www.madmode.com/2017/sel4-rpi.html>

TRENTOS-M on Raspberry Pi

- see also
 - [TRENTOS-M Hardware Platforms](#) for a general background
 - [TRENTOS-M on RPi 3](#)
 - [TRENTOS-M on RPi 4](#)
- generic platform files: https://bitbucket.hensoldt-cyber.systems/projects/SSDK/repos/plat_raspi/browse
- Demos:
 - https://bitbucket.hensoldt-cyber.systems/projects/SS/repos/demo_raspi_ethernet/browse
 - https://bitbucket.hensoldt-cyber.systems/projects/SS/repos/demo_keystore_spi_flash/browse
 - https://bitbucket.hensoldt-cyber.systems/projects/SS/repos/demo_spi_flash/browse

RasPi Sensors and Toys

- SPI temperature sensor: <https://www.amazon.de/MAX6675-Interface-Thermoelement-Temperatursensor-Arduino/dp/B01N7YW29W>
- sensor HAT: <https://www.raspberrypi.org/products/sense-hat/>
- RasPi4 compute module with NVM SSD slot:
 - <https://store.oratek.com/>
 - <https://www.tomshardware.com/news/raspberry-pi-4-compute-module-tofu>